

FAIL

High risk

The checkout load test failed, with latency and error rates exceeding the agreed limits.

Checkout load test · 5000 requests over 300s · jmeter report

WHAT THIS MEANS

One in fifty checkout attempts took more than 4 seconds, and overall two percent of requests errored, meaning shoppers could experience noticeable delays or failures during purchase. This could lead to abandoned carts and revenue loss, especially since the checkout endpoint is the critical path for conversion.

THE NUMBERS

MEDIAN	P95	P99	ERRORS	THROUGHPUT
254ms	2271ms	3565ms	2%	16.7/s

AGAINST THE AGREED THRESHOLDS

THRESHOLD	AGREED	MEASURED	OVER BY
p95 response time	2000ms	2271ms	+14%
error rate	1%	2%	+100%

KEY FINDINGS

- Overall p95 latency was 2.27 seconds, 14% above the 2 second threshold.
- Overall error rate was 2%, double the 1% limit, driven by 504 and 500 responses.
- The checkout endpoint (POST /api/checkout) had a p95 of 4.07 seconds and an error rate of 13.6%, far higher than any other endpoint.
- The slowest request recorded was 6.36 seconds, indicating occasional extreme tail latency.
- Throughput was only 16.7 requests/second, which may be below expected production load.

SLOWEST ENDPOINTS

ENDPOINT	REQUESTS	P95	ERRORS
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POST /api/checkout	612	4067ms	13.6%
GET /api/search	1056	788ms	0.1%
POST /api/cart	468	541ms	0%
POST /api/login	964	398ms	0.8%
GET /api/products	1900	281ms	0.5%

POST /api/checkout stands out with a p95 of 4.07 seconds and a 13.6% error rate, far exceeding the latency and reliability of all other endpoints.

RECOMMENDED ACTIONS

- Backend engineers should investigate the checkout endpoint, starting with logs for the 504 and 500 errors to identify timeouts or downstream failures.
- Profile the checkout service to pinpoint slow database queries or external calls that contribute to the 4-second p95 latency.
- Review and tune timeout and retry settings for services called by the checkout flow to reduce gateway (504) errors.
- After remediation, re-run the load test at production-scale traffic to verify that p95 drops below 2 seconds and error rate falls under 1%.

What this test does not prove. The test ran for 300 seconds at 16.7 RPS in a single environment, which may not reflect peak traffic patterns, longer-duration load, or production infrastructure variability. Results are limited to the measured endpoints and cannot guarantee behavior under different user mixes or network conditions.

Thresholds: p95 2000ms · p99 5000ms · error rate 1%. Every figure above was computed from the raw report; the commentary was drafted automatically and reviewed against those figures.